

UNIT IV

CORRELATION, REGRESSION AND
STATISTICAL QUALITY CONTROLCorrelation | Regression | $\bar{X}$, R , p , np and c Charts

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1 Correlation and Rank Correlation

Karl Pearson's Coefficient of Correlation (r)

$$r = \frac{n \sum xy - (\sum x)(\sum y)}{\sqrt{[n \sum x^2 - (\sum x)^2][n \sum y^2 - (\sum y)^2]}}$$

- Measures the linear relationship between two variables; $-1 \leq r \leq 1$.
- $r = +1$: perfect positive correlation; $r = -1$: perfect negative correlation; $r = 0$: no linear correlation.
- Change of origin and scale does not affect r .

Supporting summations

$$\sum x, \quad \sum y, \quad \sum x^2, \quad \sum y^2, \quad \sum xy$$

These five values are substituted directly in the correlation formula.

Spearman's Rank Correlation Coefficient (ρ)

$$\rho = 1 - \frac{6 \sum d^2}{n(n^2 - 1)}$$

- Used when the data is in rank form; $-1 \leq \rho \leq 1$.
- d = difference between the pair of ranks; n = number of ranked items.

Tied ranks adjustment

If ties occur, assign the average of the tied rank positions and use

$$\rho = 1 - \frac{6 \left[\sum d^2 + \frac{1}{12} \sum m(m^2 - 1) \right]}{n(n^2 - 1)}$$

where m is the number of tied observations in each group.

Interpretation of r and ρ

Magnitude	Interpretation
0.00 to ± 0.30	Weak correlation
± 0.30 to ± 0.70	Moderate correlation
± 0.70 to ± 1.00	Strong correlation

1.1 Problem 1 – Sales and expenditure

Problem 1

Find the coefficient of correlation between Sales (X) and Expenditure (Y) for the following 10 items.

Sales (X)	50	50	55	60	65	65	65	60	60	50
Expenditure (Y)	11	13	14	16	16	15	15	14	13	13

Solution – Table of summations

Obs.	X	Y	X^2	Y^2	XY
1	50	11	2500	121	550
2	50	13	2500	169	650
3	55	14	3025	196	770
4	60	16	3600	256	960
5	65	16	4225	256	1040
6	65	15	4225	225	975
7	65	15	4225	225	975
8	60	14	3600	196	840
9	60	13	3600	169	780
10	50	13	2500	169	650
Σ	580	140	34000	1982	8190

Substitution

$$\begin{aligned}
 \text{Numerator} &= n \sum XY - (\sum X)(\sum Y) \\
 &= 10(8190) - 580(140) = 81900 - 81200 = 700, \\
 n \sum X^2 - (\sum X)^2 &= 34000 - 336400 = 3600, \\
 n \sum Y^2 - (\sum Y)^2 &= 19820 - 19600 = 220, \\
 \text{Denominator} &= \sqrt{3600 \times 220} = \sqrt{792000} = 889.94. \\
 r &= \frac{700}{889.94} \Rightarrow \boxed{r \approx 0.79}
 \end{aligned}$$

Conclusion

$r \approx 0.79$: high positive correlation between sales and expenditure.

1.2 Problem 2 – Weight and blood pressure

Problem 2

A study has been conducted to find whether there is any relationship between the weight and the blood pressure of an individual. Determine the coefficient of correlation for the following data.

Weight X (kg)	78	86	72	82	80	88	84	89	68	71
B.P. Y (mm)	140	160	134	144	180	176	174	178	128	132

Solution – Table of summations

Obs.	X	Y	X^2	Y^2	XY
1	78	140	6084	19600	10920
2	86	160	7396	25600	13760
3	72	134	5184	17956	9648
4	82	144	6724	20736	11808
5	80	180	6400	32400	14400
6	88	176	7744	30976	15488
7	84	174	7056	30276	14616
8	89	178	7921	31684	15842
9	68	128	4624	16384	8704
10	71	132	5041	17424	9372
Σ	798	1546	64174	243036	124558

Substitution

$$\begin{aligned}
 \text{Numerator} &= 10(124558) - 798(1546) \\
 &= 1245580 - 1233708 = 11872, \\
 n \sum X^2 - (\sum X)^2 &= 641740 - 636804 = 4936, \\
 n \sum Y^2 - (\sum Y)^2 &= 2430360 - 2390116 = 40244, \\
 \text{Denominator} &= \sqrt{4936 \times 40244} = \sqrt{198644384} \\
 &= 14094.1. \\
 r &= \frac{11872}{14094.1} \Rightarrow \boxed{r \approx 0.84}
 \end{aligned}$$

Conclusion

$r \approx 0.84$: strong positive correlation between weight and blood pressure.

1.3 Problem 3 – Rank correlation (no ties)

Problem 3

Find the rank correlation coefficient for the following data.

Rank in X	1	2	3	4	5	6	7
Rank in Y	4	3	1	2	6	5	7

Solution

Rank X	1	2	3	4	5	6	7	
Rank Y	4	3	1	2	6	5	7	
$d = X - Y$	-3	-1	2	2	-1	1	0	
d^2	9	1	4	4	1	1	0	$\sum d^2 = 20$

With $n = 7$,

$$\rho = 1 - \frac{6(20)}{7(7^2 - 1)} = 1 - \frac{120}{336} = 1 - \frac{5}{14} \Rightarrow \boxed{\rho = \frac{9}{14} \approx 0.64}$$

1.4 Problem 4 – Rank correlation with tied ranks

Problem 4

Find the rank correlation coefficient for the following marks.

Marks in X	68	84	75	50	64	80	75	40	55	64
Marks in Y	62	58	68	45	81	60	68	48	50	70

Assigning ranks (smallest mark = rank 1; ties get the average rank)

Sorted X : 40, 50, 55, 64, 64, 68, 75, 75, 80, 84 $\Rightarrow$ 40 $\rightarrow$ 1, 50 $\rightarrow$ 2, 55 $\rightarrow$ 3, 64, 64 $\rightarrow \frac{4+5}{2} = 4.5$, 68 $\rightarrow$ 6, 75, 75 $\rightarrow \frac{7+8}{2} = 7.5$, 80 $\rightarrow$ 9, 84 $\rightarrow$ 10.

Sorted Y : 45, 48, 50, 58, 60, 62, 68, 68, 70, 81 $\Rightarrow$ 45 $\rightarrow$ 1, 48 $\rightarrow$ 2, 50 $\rightarrow$ 3, 58 $\rightarrow$ 4, 60 $\rightarrow$ 5, 62 $\rightarrow$ 6, 68, 68 $\rightarrow 7.5$, 70 $\rightarrow$ 9, 81 $\rightarrow$ 10.

Solution

Item	1	2	3	4	5	6	7	8	9	10	
R_X	6	10	7.5	2	4.5	9	7.5	1	3	4.5	
R_Y	6	4	7.5	1	10	5	7.5	2	3	9	
d	0	6	0	1	-5.5	4	0	-1	0	-4.5	
d^2	0	36	0	1	30.25	16	0	1	0	20.25	$\sum d^2 = 104.5$

Tie correction

There are three tied groups, each of size $m = 2$ (the two 64's and the two 75's in X , and the two 68's in Y), so

$$\frac{1}{12} \sum m(m^2 - 1) = \frac{1}{12} [2(3) + 2(3) + 2(3)] = \frac{18}{12} = 1.5.$$

$$\rho = 1 - \frac{6[\sum d^2 + 1.5]}{n(n^2 - 1)} = 1 - \frac{6(104.5 + 1.5)}{10(99)} = 1 - \frac{636}{990} \Rightarrow \boxed{\rho \approx 0.36}$$

Note

If the tie correction is omitted, $\rho = 1 - \frac{6(104.5)}{990} \approx 0.37$. Since ties are present, the corrected value $\rho \approx 0.36$ is the proper one. Either way the association is **low positive**.

1.5 Problem 5 – Musical contest rankings**Problem 5**

Ten competitors in a musical contest were ranked by three judges A , B , C in the following order. Using Spearman's rank correlation, find which pair of judges have the same liking in music.

Rank by A	1	6	5	10	3	2	4	9	7	8
Rank by B	3	5	8	6	4	7	10	2	1	9
Rank by C	4	6	9	8	1	2	3	10	5	7

(i) Judges A and B

A	1	6	5	10	3	2	4	9	7	8	
B	3	5	8	6	4	7	10	2	1	9	
d^2	4	1	9	16	1	25	36	49	36	1	$\sum d^2 = 178$

$$\rho_{AB} = 1 - \frac{6(178)}{990} = 1 - \frac{1068}{990} \Rightarrow \boxed{\rho_{AB} \approx -0.08}$$

(ii) Judges A and C

A	1	6	5	10	3	2	4	9	7	8	
C	4	6	9	8	1	2	3	10	5	7	
d^2	9	0	16	4	4	0	1	1	4	1	$\sum d^2 = 40$

$$\rho_{AC} = 1 - \frac{6(40)}{990} = 1 - \frac{8}{33} \Rightarrow \boxed{\rho_{AC} = \frac{25}{33} \approx 0.76}$$

(iii) Judges *B* and *C*

<i>B</i>	3	5	8	6	4	7	10	2	1	9	
<i>C</i>	4	6	9	8	1	2	3	10	5	7	
d^2	1	1	1	4	9	25	49	64	16	4	$\sum d^2 = 174$

$$\rho_{BC} = 1 - \frac{6(174)}{990} = 1 - \frac{1044}{990} \Rightarrow \boxed{\rho_{BC} \approx -0.05}$$

Conclusion

$$\rho_{AB} \approx -0.08, \quad \rho_{AC} \approx 0.76, \quad \rho_{BC} \approx -0.05$$

The highest (strongly positive) rank correlation is between *A* and *C*.

Judges *A* and *C* have the same liking in music.

2 Regression Lines

Regression coefficients from the raw sums

$$\text{Regression of } Y \text{ on } X: Y - \bar{Y} = b_{Y|X}(X - \bar{X}), \quad b_{Y|X} = \frac{n \sum xy - (\sum x)(\sum y)}{n \sum x^2 - (\sum x)^2}$$

$$\text{Regression of } X \text{ on } Y: X - \bar{X} = b_{X|Y}(Y - \bar{Y}), \quad b_{X|Y} = \frac{n \sum xy - (\sum x)(\sum y)}{n \sum y^2 - (\sum y)^2}$$

$$\boxed{b_{Y|X} b_{X|Y} = r^2} \quad \text{and} \quad b_{Y|X} = r \frac{\sigma_Y}{\sigma_X}, \quad b_{X|Y} = r \frac{\sigma_X}{\sigma_Y}$$

How to identify which line is which

Of the two given lines, the assignment of $b_{Y|X}$ and $b_{X|Y}$ must satisfy $b_{Y|X} b_{X|Y} = r^2 \leq 1$. If a choice gives $r^2 > 1$, the other assignment is the correct one. Both coefficients always carry the same sign, which is the sign of r .

2.1 Problem 1 – Engine age and maintenance cost

Problem 1

The data below show engine age x (years) and annual maintenance cost y (Rs. $\times 1000$). Find (i) the regression equation of Y on X , and (ii) the predicted maintenance cost for a 4-year-old engine.

Age (years)	1	3	5	7	9
Maintenance cost (Rs. $\times 1000$)	15	18	21	23	22

Solution

$$n = 5, \quad \sum x = 25, \quad \sum y = 99, \quad \sum x^2 = 165, \quad \sum xy = 533$$

$$\bar{x} = 5, \quad \bar{y} = 19.8$$

$$b_{Y|X} = \frac{5(533) - 25(99)}{5(165) - 25^2} = \frac{2665 - 2475}{825 - 625} = \frac{190}{200} \Rightarrow \boxed{b_{Y|X} = 0.95}$$

$$y - 19.8 = 0.95(x - 5) \Rightarrow \boxed{y = 0.95x + 15.05}$$

$$\hat{y}|_{x=4} = 0.95(4) + 15.05 \Rightarrow \boxed{\hat{y} = 18.85}$$

Conclusion

Regression line $y = 0.95x + 15.05$; the predicted maintenance cost of a 4-year-old engine is **Rs. 18 850**.

2.2 Problem 2 – Engine speed and fuel consumption**Problem 2**

Given $\text{Var}(X) = 25$ and the two regression lines $5x - y = 22$ and $64x - 45y = 24$, find the means $\bar{x}, \bar{y}$, the correlation r and σ_Y .

Means

The two lines intersect at $(\bar{x}, \bar{y})$:

$$5\bar{x} - \bar{y} = 22, \quad 64\bar{x} - 45\bar{y} = 24$$

$$\bar{y} = 5\bar{x} - 22 \Rightarrow 64\bar{x} - 45(5\bar{x} - 22) = 24 \Rightarrow -161\bar{x} = -966$$

$$\boxed{\bar{x} = 6}$$

$$\boxed{\bar{y} = 8}$$

Correlation coefficient

Taking $5x - y = 22$ as the line of X on Y and $64x - 45y = 24$ as the line of Y on X ,

$$b_{X|Y} = \frac{1}{5}, \quad b_{Y|X} = \frac{64}{45}$$

$$r^2 = b_{X|Y} b_{Y|X} = \frac{1}{5} \cdot \frac{64}{45} = \frac{64}{225} = \left(\frac{8}{15}\right)^2 \Rightarrow \boxed{r = \frac{8}{15} \approx 0.53}$$

(The opposite assignment would give $r^2 = \frac{225}{64} > 1$, which is impossible.)

Standard deviation of Y

$$\sigma_X = \sqrt{25} = 5 \text{ and } b_{Y|X} = r \frac{\sigma_Y}{\sigma_X}, \text{ so}$$

$$\frac{64}{45} = \frac{8}{15} \cdot \frac{\sigma_Y}{5} \quad \Rightarrow \quad \boxed{\sigma_Y = \frac{40}{3} \approx 13.33}$$

2.3 Problem 3 – Heat transfer and flow rate**Problem 3**

Given $8Q - 10F + 66 = 0$, $40Q - 18F - 214 = 0$ and $\text{Var}(Q) = 9$, find the mean values, r and σ_F .

Means

$$8\bar{Q} - 10\bar{F} = -66, \quad 40\bar{Q} - 18\bar{F} = 214$$

Multiplying the first by 5 and subtracting the second: $-32\bar{F} = -544$, so

$$\boxed{\bar{F} = 17} \quad \text{and} \quad 8\bar{Q} = 10(17) - 66 = 104 \quad \Rightarrow \quad \boxed{\bar{Q} = 13}$$

Correlation coefficient and σ_F

$$b_{F|Q} = \frac{8}{10} = \frac{4}{5}, \quad b_{Q|F} = \frac{18}{40} = \frac{9}{20}$$

$$r^2 = \frac{4}{5} \cdot \frac{9}{20} = \frac{9}{25} = \left(\frac{3}{5}\right)^2 \quad \Rightarrow \quad \boxed{r = 0.6}$$

$$\text{With } \sigma_Q = \sqrt{9} = 3 \text{ and } b_{F|Q} = r \frac{\sigma_F}{\sigma_Q},$$

$$\frac{4}{5} = \frac{3}{5} \cdot \frac{\sigma_F}{3} \quad \Rightarrow \quad \boxed{\sigma_F = 4}$$

3 Statistical Quality Control – Introduction**What is SQC?**

Statistical Quality Control (SQC) is a method used in the manufacturing and service industries to monitor and control the quality of a product or process. It uses statistical techniques to ensure that a product or process meets the desired specifications and standards. One of the most common tools in SQC is the **control chart**.

Types of control charts – 1. For variables

For characteristics measured on a continuous scale (weight, length, temperature, ...):

- $\bar{X}$ -chart (mean chart) – monitors the average of a process over time.
- R -chart (range chart) – monitors the variability of the process.

Types of control charts – 2. For attributes

For characteristics that are counted rather than measured:

- p -chart – monitors the proportion of defective units in a sample.
- np -chart – monitors the number of defective units in a sample.
- c -chart – monitors the number of defects per unit.

Reading any control chart

- If all points lie within the control limits and show no pattern, the process is **in control**.
- If any point lies outside the limits, or a non-random pattern (such as a trend) is present, the process may be **out of control**, indicating special causes of variation that need investigation.

Control chart constants used in this unit

Sample size n	A_2	D_3	D_4
3	1.023	0	2.574
5	0.577	0	2.114
6	0.483	0	2.004

4 $\bar{X}$ -Chart (Mean Chart)**Steps in constructing an $\bar{X}$ -chart**

1. Collect k samples, each of size n .
2. Compute the sample mean $\bar{X}_i$ and the range R_i for each sample.
3. Compute the grand mean $\bar{\bar{X}} = \frac{\sum_{i=1}^k \bar{X}_i}{k}$ and the average range $\bar{R} = \frac{\sum_{i=1}^k R_i}{k}$.
4. Compute the control limits.
5. Plot the sample means together with CL, UCL and LCL, and interpret.

Control limits for the $\bar{X}$ -chart

$$CL = \bar{\bar{X}}, \quad UCL = \bar{\bar{X}} + A_2\bar{R}, \quad LCL = \bar{\bar{X}} - A_2\bar{R}$$

where A_2 depends on the sample size n and $\bar{R}$ is the average range.

4.1 Problem 1 – Diameter of bearings**Problem 1**

A factory produces bearings, and the diameter of the bearings is critical to their performance. The factory collects data on the diameter of 5 bearings every hour for 10 hours. Check whether the process is under control.

Sample	Diameter (mm)	$\bar{X}_i$	R_i
1	10.01, 10.03, 9.99, 10.00, 10.02	10.010	0.04
2	10.05, 10.06, 10.04, 10.03, 10.01	10.038	0.05
3	9.98, 9.97, 9.99, 10.01, 10.02	9.994	0.05
4	10.00, 10.01, 9.99, 10.00, 10.03	10.006	0.04
5	10.02, 10.00, 10.01, 9.99, 10.00	10.004	0.03
6	9.99, 10.00, 9.98, 10.02, 9.99	9.996	0.04
7	10.01, 10.03, 10.00, 10.02, 9.99	10.010	0.04
8	10.03, 10.04, 10.02, 10.00, 10.01	10.020	0.04
9	10.00, 9.99, 10.01, 10.02, 9.98	10.000	0.04
10	9.97, 9.98, 10.00, 10.01, 10.02	9.996	0.05

Grand mean and average range

$$\begin{aligned} \bar{\bar{X}} &= \frac{\sum \bar{X}_i}{k} = \frac{10.010 + 10.038 + 9.994 + 10.006 + 10.004}{10} \\ &\quad + \frac{9.996 + 10.010 + 10.020 + 10.000 + 9.996}{10} \\ &= \frac{100.074}{10} \Rightarrow \boxed{\bar{\bar{X}} = 10.0074} \end{aligned}$$

$$\begin{aligned} \bar{R} &= \frac{0.04 + 0.05 + 0.05 + 0.04 + 0.03 + 0.04 + 0.04 + 0.04 + 0.04 + 0.05}{10} \\ &= \frac{0.42}{10} \Rightarrow \boxed{\bar{R} = 0.042} \end{aligned}$$

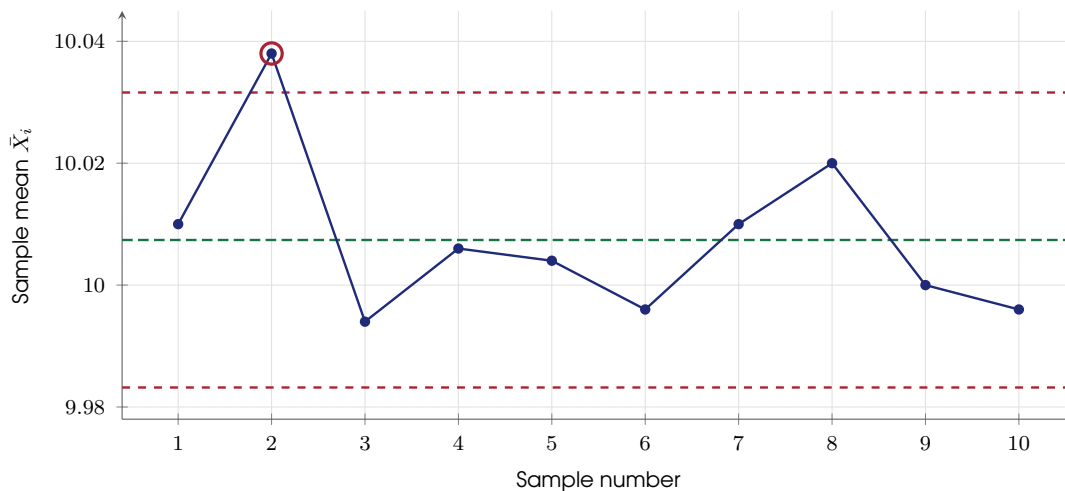
Control limits (for $n = 5$, $A_2 = 0.577$)

$$A_2 \bar{R} = (0.577)(0.042) = 0.0242$$

$$UCL = \bar{\bar{X}} + A_2 \bar{R} = 10.0074 + 0.0242 \Rightarrow \boxed{UCL = 10.032}$$

$$CL = \boxed{10.007}$$

$$LCL = \bar{\bar{X}} - A_2 \bar{R} = 10.0074 - 0.0242 \Rightarrow \boxed{LCL = 9.983}$$

**Conclusion – the process is NOT in control**

The mean of sample 2 is $\bar{X}_2 = 10.038$, which lies **above** the upper control limit $UCL = 10.032$ (circled on the chart). All the other nine sample means lie within the limits.

Since one point falls outside the control limits, the process is out of control; sample 2 should be investigated for an assignable (special) cause.

4.2 Problem 2 – Thickness of metal sheets**Problem 2**

A manufacturer monitors the average thickness of metal sheets. Every 30 minutes the thickness of 6 metal sheets is measured. Check whether the process is under control.

Sample	Thickness (mm)	$\bar{X}_i$	R_i
1	1.02, 1.00, 1.01, 1.03, 1.01, 1.02	1.015	0.03
2	1.04, 1.02, 1.01, 1.03, 1.02, 1.05	1.028	0.04
3	1.01, 1.00, 1.03, 1.02, 1.04, 1.01	1.018	0.04
4	1.02, 1.03, 1.01, 1.04, 1.00, 1.02	1.020	0.04
5	1.03, 1.01, 1.02, 1.03, 1.05, 1.04	1.030	0.04
6	1.02, 1.00, 1.01, 1.03, 1.04, 1.05	1.025	0.05
7	1.01, 1.03, 1.00, 1.02, 1.01, 1.04	1.018	0.04
8	1.04, 1.02, 1.01, 1.05, 1.03, 1.02	1.028	0.04

Grand mean and average range

$$\bar{\bar{X}} = \frac{1.015 + 1.028 + 1.018 + 1.020 + 1.030 + 1.025 + 1.018 + 1.028}{8}$$

$$= \frac{8.182}{8} \Rightarrow \boxed{\bar{\bar{X}} = 1.023}$$

$$\bar{R} = \frac{0.03 + 0.04 + 0.04 + 0.04 + 0.04 + 0.05 + 0.04 + 0.04}{8}$$

$$= \frac{0.32}{8} \Rightarrow \boxed{\bar{R} = 0.040}$$

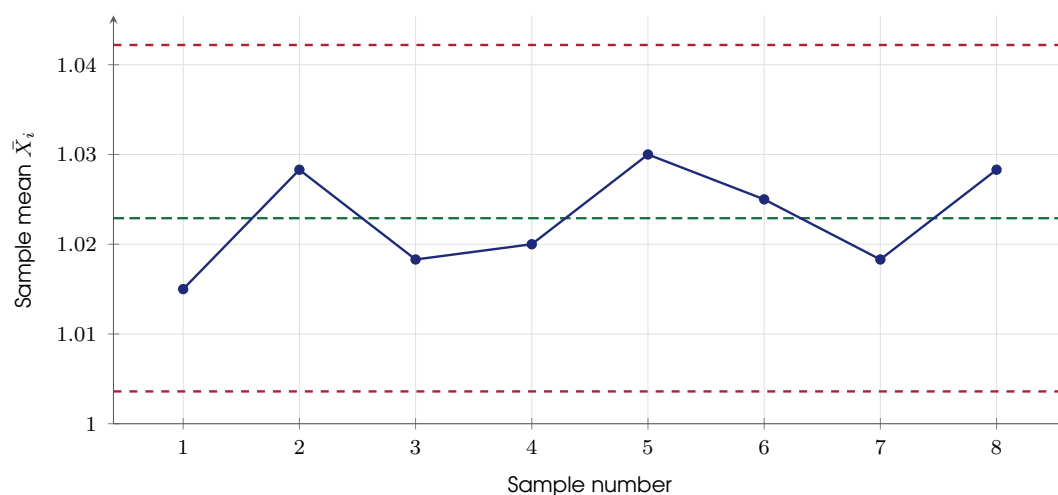
Control limits (for $n = 6$, $A_2 = 0.483$)

$$A_2 \bar{R} = (0.483)(0.040) = 0.019$$

$$UCL = \bar{\bar{X}} + A_2 \bar{R} = 1.023 + 0.019 \Rightarrow \boxed{UCL = 1.042}$$

$$CL = \boxed{1.023}$$

$$LCL = \bar{\bar{X}} - A_2 \bar{R} = 1.023 - 0.019 \Rightarrow \boxed{LCL = 1.004}$$



Conclusion

All eight sample means lie between $LCL = 1.004$ and $UCL = 1.042$, with no trend or pattern.

The process is in control.

5 R -Chart (Range Chart)

What an R -chart does

An R -chart monitors the *variability* or dispersion of a process over time. It plots the range (the difference between the largest and the smallest value) of each sample. The R -chart is normally used together with the $\bar{X}$ -chart, so that both the process average and the process variability are monitored.

Components

1. **Centre line (CL)** – the average range $\bar{R}$ of the subgroups.
2. **Upper control limit (UCL)** – points above this line indicate an out-of-control process.
3. **Lower control limit (LCL)** – points below this line may indicate an unusual decrease in process variation.

Control limits for the R -chart

$$CL = \bar{R}, \quad UCL = D_4 \bar{R}, \quad LCL = D_3 \bar{R}$$

where D_3 and D_4 are constants depending on the sample size n .

5.1 Problem 1 – Diameter of steel rods

Problem 1

A factory is monitoring the consistency of the diameter of steel rods. Five samples are taken every hour, and the results from six consecutive hours are given below (diameter in mm).

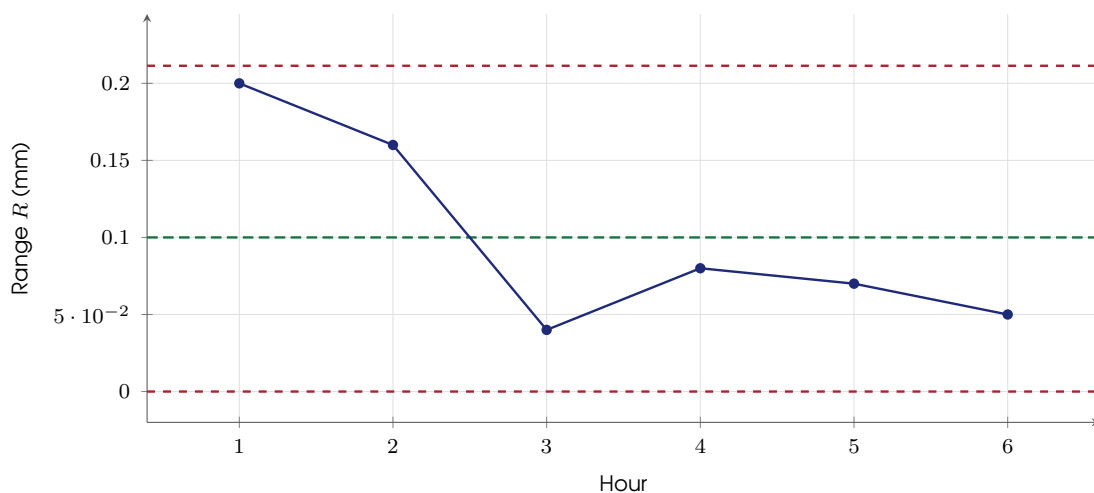
Hour	Sample 1	Sample 2	Sample 3	Sample 4	Sample 5	Range R
1	4.90	5.10	5.00	5.05	5.02	0.20
2	4.92	5.08	5.01	5.04	5.00	0.16
3	5.01	5.04	5.02	5.03	5.00	0.04
4	4.98	5.06	5.01	5.02	5.03	0.08
5	5.02	5.07	5.03	5.00	5.01	0.07
6	5.00	5.02	5.03	5.05	5.01	0.05

Average range and control limits ($n = 5: D_3 = 0, D_4 = 2.114$)

$$\bar{R} = \frac{0.20 + 0.16 + 0.04 + 0.08 + 0.07 + 0.05}{6} = \frac{0.60}{6} \Rightarrow \boxed{\bar{R} = 0.10}$$

$$UCL = D_4 \bar{R} = (2.114)(0.10) \Rightarrow \boxed{UCL = 0.2114}$$

$$LCL = D_3 \bar{R} = (0)(0.10) \Rightarrow \boxed{LCL = 0}$$



Conclusion

No point falls outside the control limits, so there are no signs of special-cause variation.

The process variability is stable over time.

5.2 Problem 2 – Temperature in a chemical process

Problem 2

In a chemical process the temperature is monitored to maintain process stability. Three samples are taken at four different time intervals, with the temperatures (in °C) below. Check the variability of the process.

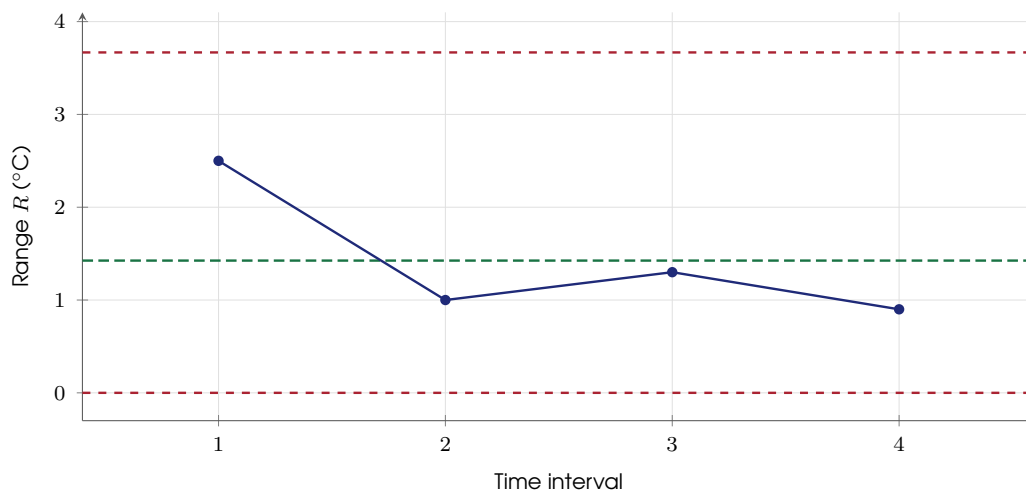
Time interval	Sample 1	Sample 2	Sample 3	Range R
1	78.0	80.5	79.5	2.5
2	79.0	79.2	80.0	1.0
3	78.5	79.8	79.3	1.3
4	79.1	79.7	78.8	0.9

Average range and control limits ($n = 3$: $D_3 = 0$, $D_4 = 2.574$)

$$\bar{R} = \frac{2.5 + 1.0 + 1.3 + 0.9}{4} = \frac{5.7}{4} \Rightarrow \boxed{\bar{R} = 1.425}$$

$$UCL = (2.574)(1.425) \Rightarrow \boxed{UCL = 3.668}$$

$$LCL = (0)(1.425) \Rightarrow \boxed{LCL = 0}$$



Conclusion

All the ranges fall within the control limits.

The temperature variation is under control and the process is stable.

6 p -Chart (Proportion Chart)

What a p -chart does

p -charts, or proportion charts, are used to monitor the proportion of defective items in a process. They are ideal for attribute data (pass/fail, defective/non-defective, yes/no). The chart plots the proportion of defective items in each sample over time and helps to identify variations in the process.

Control limits for the p -chart

$$CL = \bar{p} = \frac{\text{Total defective items}}{\text{Total sample size}}$$

$$UCL = \bar{p} + 3\sqrt{\frac{\bar{p}(1-\bar{p})}{n}}, \quad LCL = \bar{p} - 3\sqrt{\frac{\bar{p}(1-\bar{p})}{n}}$$

where n is the sample size. If LCL is negative it is set to zero, because a negative proportion of defectives has no meaning.

6.1 Problem 1 – Defective units in batches of 200

Problem 1

A factory produces batches of 200 units each. The management wants to monitor the proportion of defective units over 10 consecutive days. Determine whether the process is under control.

Day	1	2	3	4	5	6	7	8	9	10
Defectives	6	4	5	9	3	5	7	6	4	8

Proportion defective for each day ($p_i = \frac{\text{defectives}}{200}$)

Day	1	2	3	4	5	6	7	8	9	10
p_i	0.030	0.020	0.025	0.045	0.015	0.025	0.035	0.030	0.020	0.040

$$\bar{p} = \frac{\text{Total defectives}}{\text{Total units}} = \frac{57}{2000} \Rightarrow \boxed{\bar{p} = 0.0285}$$

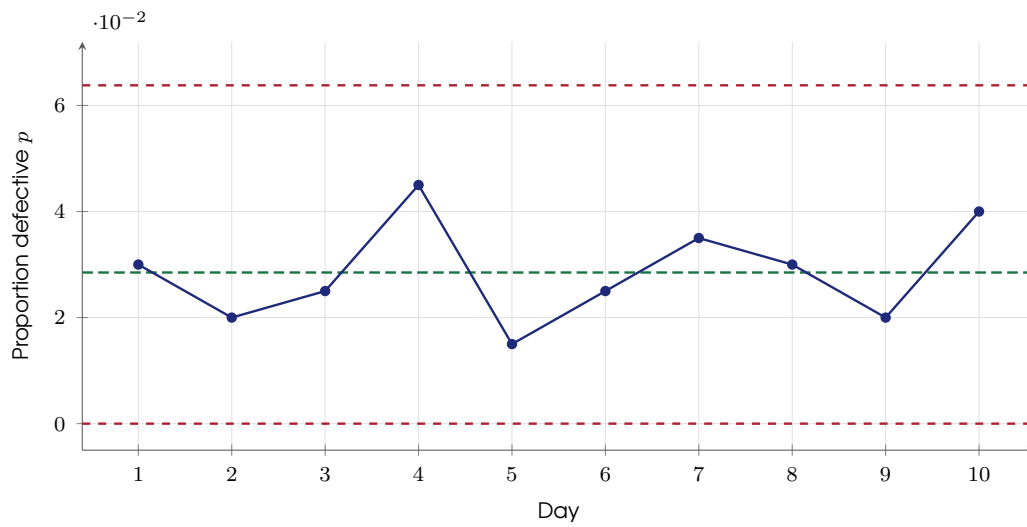
Control limits ($n = 200$)

$$3\sqrt{\frac{\bar{p}(1-\bar{p})}{n}} = 3\sqrt{\frac{0.0285(1-0.0285)}{200}} = 3\sqrt{\frac{0.0276878}{200}}$$

$$= 3(0.011766) = 0.0353$$

$$UCL = 0.0285 + 0.0353 \Rightarrow \boxed{UCL = 0.0638}$$

$$LCL = 0.0285 - 0.0353 = -0.0068 \Rightarrow \boxed{LCL = 0}$$

**Conclusion**

All ten proportions lie between $LCL = 0$ and $UCL = 0.0638$.

The process is under control.

6.2 Problem 2 – Errors in insurance claims**Problem 2**

An insurance company reviews 100 claims per day over 12 days to monitor errors in processing. The number of claims with errors per day is shown below. Check whether the process is in control.

Day	1	2	3	4	5	6	7	8	9	10	11	12
Errors	3	1	2	4	3	2	5	3	1	4	3	2

Proportion defective and $\bar{p}$

Day	1	2	3	4	5	6	7	8	9	10	11	12
p_i	0.03	0.01	0.02	0.04	0.03	0.02	0.05	0.03	0.01	0.04	0.03	0.02

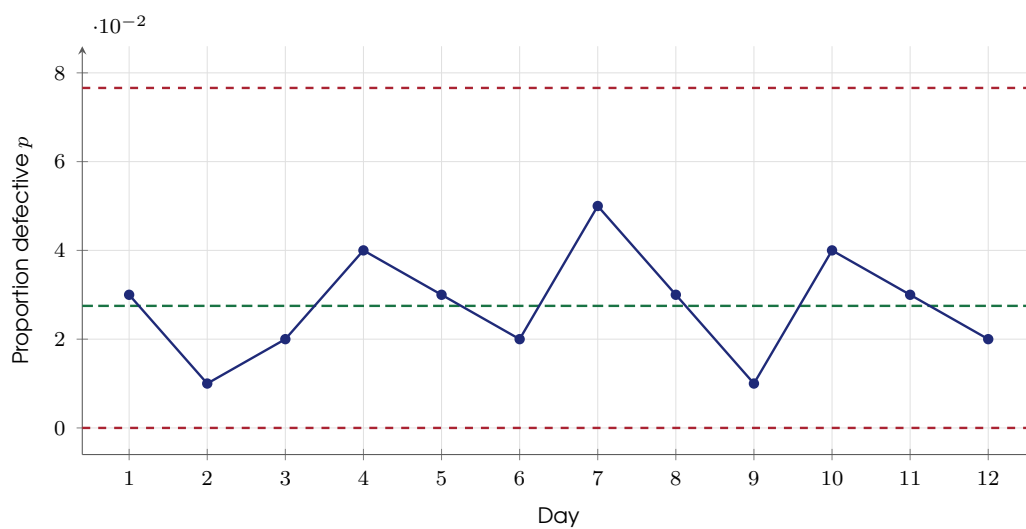
$$\bar{p} = \frac{33}{1200} \Rightarrow \boxed{\bar{p} = 0.0275}$$

Control limits ($n = 100$)

$$3\sqrt{\frac{\bar{p}(1-\bar{p})}{n}} = 3\sqrt{\frac{0.0275(0.9725)}{100}} = 3(0.016354) = 0.0491$$

$$UCL = 0.0275 + 0.0491 \Rightarrow \boxed{UCL = 0.0766}$$

$$LCL = 0.0275 - 0.0491 = -0.0216 \Rightarrow \boxed{LCL = 0}$$



Conclusion

No point exceeds the control limits.
The process appears to be stable and in control.

7 np -Chart (Number of Defectives)

What an np -chart does

- np -charts are used when monitoring the actual *number* of defective items rather than the proportion.
- They are best used when the sample size is constant.
- The chart shows the number of defective items in each sample and helps in identifying variations in the process.

Control limits for the np -chart

$CL = n\bar{p}$, $UCL = n\bar{p} + 3\sqrt{n\bar{p}(1-\bar{p})}$, $LCL = n\bar{p} - 3\sqrt{n\bar{p}(1-\bar{p})}$
 where $\bar{p} = \frac{n\bar{p}}{n}$ and n is the sample size. If LCL is negative it is set to 0, as the number of defectives cannot be negative.

7.1 Problem 1 – Defective light bulbs**Problem 1**

A factory produces light bulbs in batches of 150. The number of defective bulbs per batch is recorded over 8 days. Construct an np -chart and check the process.

Day	1	2	3	4	5	6	7	8
Defectives	7	5	6	10	4	8	6	9

Average number of defectives

$$n\bar{p} = \frac{7 + 5 + 6 + 10 + 4 + 8 + 6 + 9}{8} = \frac{55}{8} \Rightarrow \boxed{n\bar{p} = 6.875}$$

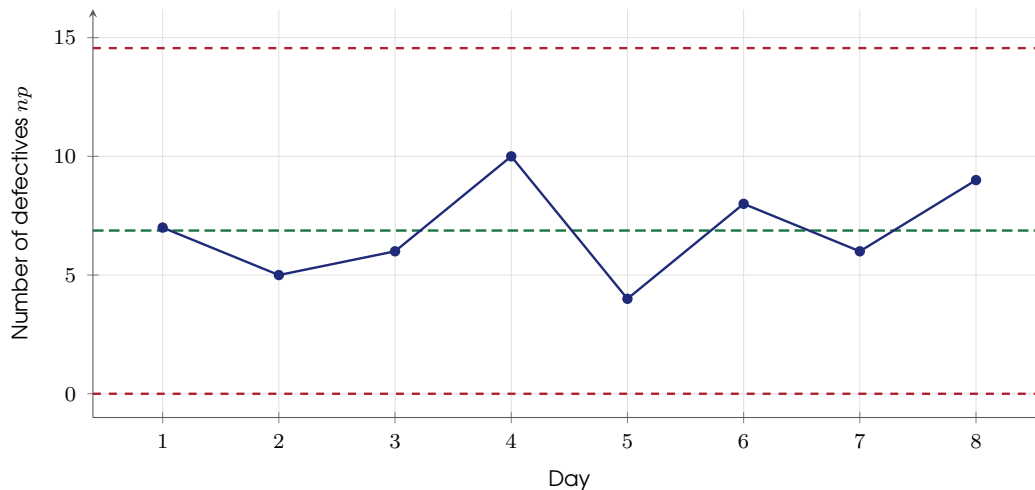
$$\bar{p} = \frac{n\bar{p}}{n} = \frac{6.875}{150} \Rightarrow \boxed{\bar{p} = 0.04583}$$

Control limits ($n = 150$)

$$\begin{aligned}
 3\sqrt{n\bar{p}(1-\bar{p})} &= 3\sqrt{6.875(1-0.04583)} = 3\sqrt{6.875 \times 0.95417} \\
 &= 3\sqrt{6.5599} = 3(2.5613) = 7.684
 \end{aligned}$$

$$UCL = 6.875 + 7.684 \Rightarrow \boxed{UCL = 14.559}$$

$$LCL = 6.875 - 7.684 = -0.809 \Rightarrow \boxed{LCL = 0}$$

**Conclusion**

All eight points lie within the control limits.
The process remains within control limits, indicating stability.

7.2 Problem 2 – Printing errors**Problem 2**

A printing press examines 200 pages daily for printing errors. The number of pages with errors over 6 days is recorded below. Construct an np -chart.

Day	1	2	3	4	5	6
Errors	5	3	4	8	2	6

Average number of defectives and control limits ($n = 200$)

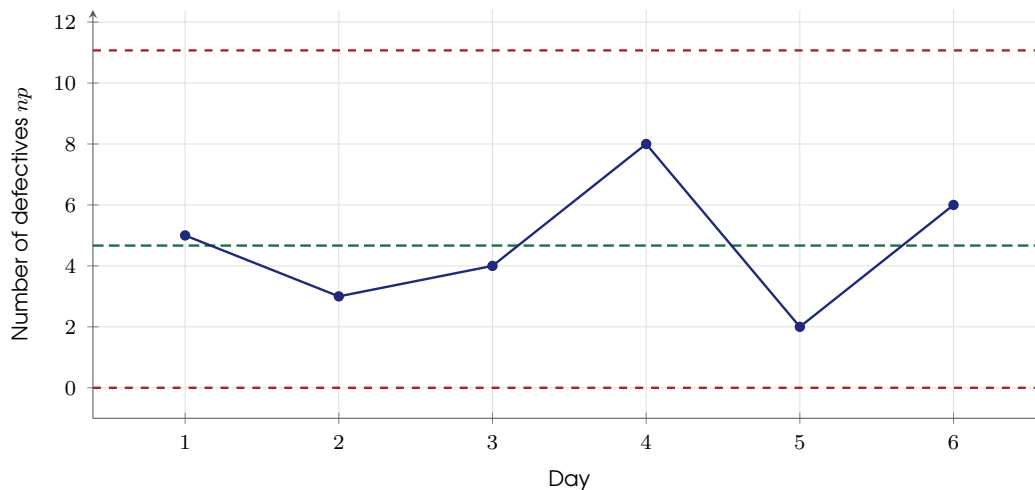
$$n\bar{p} = \frac{5 + 3 + 4 + 8 + 2 + 6}{6} = \frac{28}{6} \Rightarrow \boxed{n\bar{p} = 4.667}$$

$$\bar{p} = \frac{4.667}{200} \Rightarrow \boxed{\bar{p} = 0.02333}$$

$$3\sqrt{n\bar{p}(1 - \bar{p})} = 3\sqrt{4.667 \times 0.97667} = 3\sqrt{4.5581} = 3(2.1350) = 6.405$$

$$UCL = 4.667 + 6.405 \Rightarrow \boxed{UCL = 11.072}$$

$$LCL = 4.667 - 6.405 = -1.738 \Rightarrow \boxed{LCL = 0}$$



Conclusion

All six points lie within the control limits.

The process remains within control limits, indicating stability.

8 c -Chart (Control Chart for Count of Defects)

What a c -chart does

A c -chart is used for monitoring the count of defects or nonconformities in a process when the sample size remains constant. It is useful for tracking the number of defects in manufacturing parts, services or processes over time. The c -chart is based on the **Poisson distribution**, which is suitable for count data.

Key components

1. **Centre line (CL)** – the average number of defects per unit.
2. **Upper control limit (UCL)** – points above this limit indicate that the process is out of control.
3. **Lower control limit (LCL)** – points below this limit may also signal an issue, such as miscounts or process changes.

Control limits for the c -chart

$$CL = \bar{c} = \frac{\sum c_i}{k}, \quad UCL = \bar{c} + 3\sqrt{\bar{c}}, \quad LCL = \bar{c} - 3\sqrt{\bar{c}}$$

where k is the number of inspections. If LCL is negative it is set to 0.

8.1 Problem 1 – Defects in car seats

Problem 1

A company manufactures car seats, and quality control is done by inspecting 100 car seats every day for defects. The defects observed in the last 10 days are given below. Are the defects within acceptable control limits?

Day	1	2	3	4	5	6	7	8	9	10
Defects	5	3	6	4	7	3	4	8	5	4

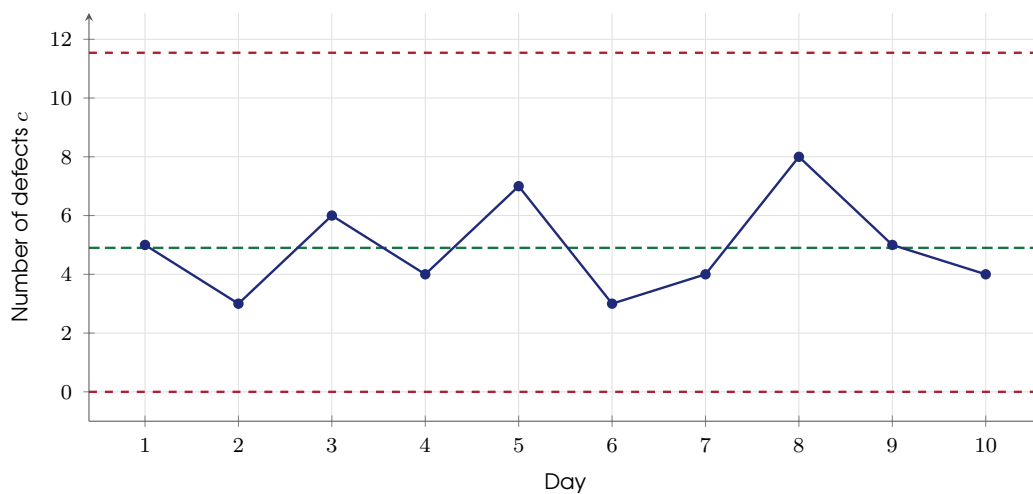
Average number of defects and control limits

$$\bar{c} = \frac{5 + 3 + 6 + 4 + 7 + 3 + 4 + 8 + 5 + 4}{10} = \frac{49}{10} \Rightarrow \boxed{\bar{c} = 4.9}$$

$$3\sqrt{\bar{c}} = 3\sqrt{4.9} = 3(2.2136) = 6.641$$

$$UCL = \bar{c} + 3\sqrt{\bar{c}} = 4.9 + 6.641 \Rightarrow \boxed{UCL = 11.54}$$

$$LCL = \bar{c} - 3\sqrt{\bar{c}} = 4.9 - 6.641 = -1.741 \Rightarrow \boxed{LCL = 0}$$



Conclusion

The defects are mostly below the centre line ($CL = 4.9$) and none of the points exceeds $UCL = 11.54$.

The process is in control, with no signs of special-cause variation.

8.2 Problem 2 – Defects in a printing process

Problem 2

A printing company inspects sheets of printed material for defects. In each inspection 50 sheets are examined. The number of defects observed over 12 consecutive inspections is given below. Are the defects within acceptable control limits?

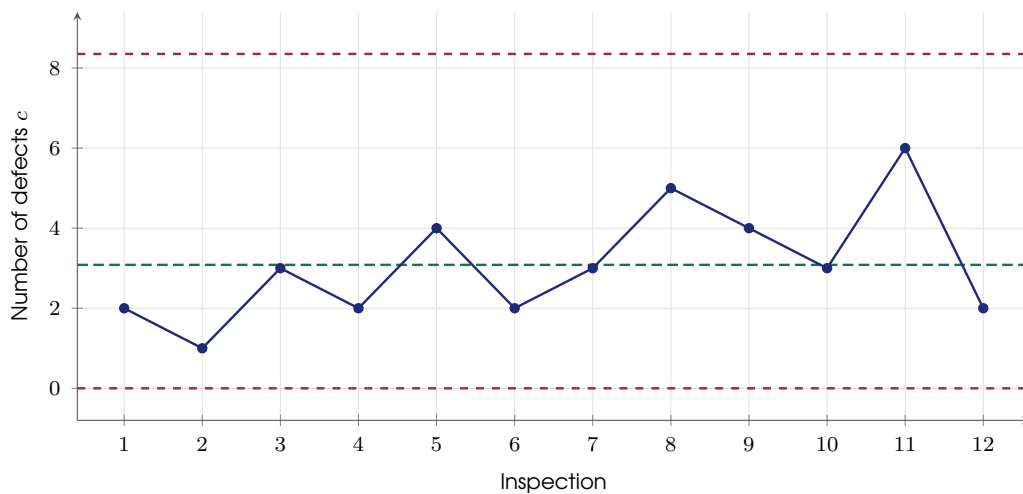
Inspection	1	2	3	4	5	6	7	8	9	10	11	12
Defects	2	1	3	2	4	2	3	5	4	3	6	2

Average number of defects and control limits

$$\bar{c} = \frac{2 + 1 + 3 + 2 + 4 + 2 + 3 + 5 + 4 + 3 + 6 + 2}{12} = \frac{37}{12} \Rightarrow \boxed{\bar{c} = 3.08}$$

$$UCL = 3.08 + 3\sqrt{3.08} = 3.08 + 3(1.7550) = 3.08 + 5.265 \Rightarrow \boxed{UCL = 8.35}$$

$$LCL = 3.08 - 5.265 = -2.185 \Rightarrow \boxed{LCL = 0}$$



Conclusion

The number of defects fluctuates about the centre line ($CL = 3.08$) with no point exceeding $UCL = 8.35$.

The printing process is in control.

8.3 Exercise – Monitoring defects in a bottling process

Exercise

A beverage company is monitoring the defects in its bottling process. Every day the company inspects 50 bottles and counts the number of defects, such as improper caps, labels or damaged bottles. The defects observed over 10 consecutive days are recorded below.

Day	1	2	3	4	5	6	7	8	9	10
Defects	3	4	2	5	3	6	7	3	4	2

Construct the $\bar{c}$ -chart to monitor the number of defects over the 10-day period and identify any points that indicate the process is out of control.

Answer

$$\bar{c} = \frac{39}{10} = 3.9, \quad UCL = 3.9 + 3\sqrt{3.9} = 9.82, \quad LCL = 0.$$

All ten points lie within the limits, so the bottling process is in control.

* * *